R26-IT-130

LexiLand

A Bilingual Gamified Platform for Dyslexia Screening and personalized Learning Support for Sri Lankan Primary Students

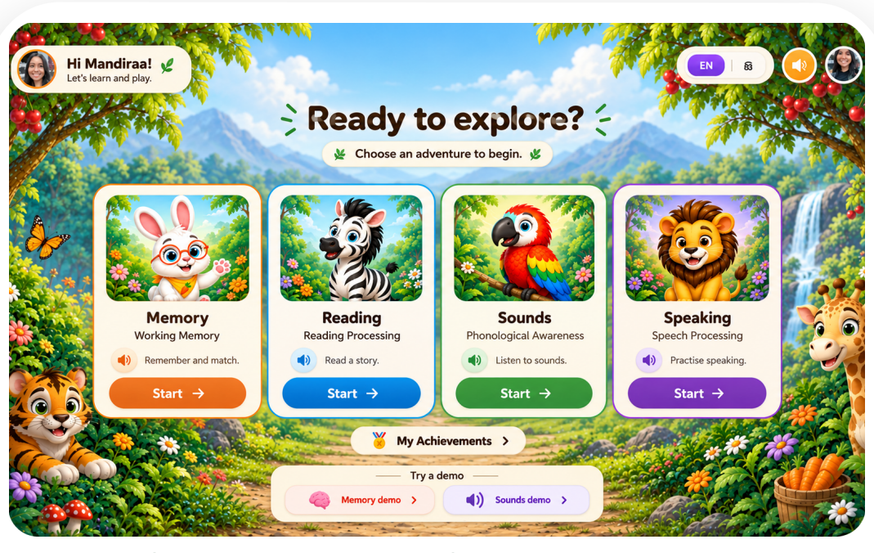
Ages 7-9 · Grades 2-4 · Sinhala & English

Dewminie W.T.M[1], Fernando K.H.D.M[1], Khan N.A.B[1], Navod J.N[1], Sanvitha Kasthuriarachchi[2], Thisara Shyamalee[2]

[1]Department of Information Technology, Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka
[2]First Year Division, Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka

INTRODUCTION

Dyslexia-related difficulties in memory, sound awareness, reading and speech often go unidentified in Sri Lankan primary schools, and few engaging tools exist in Sinhala. Most existing apps are English-only and target a single difficulty. **LexiLand** is a functional, bilingual web platform that combines educational screening with **personalized games** for children aged 7-9. It uses **Random Forest**, **MediaPipe** gaze tracking and **Whisper** speech recognition, with audio guidance, animal guides and **color-coded levels**.



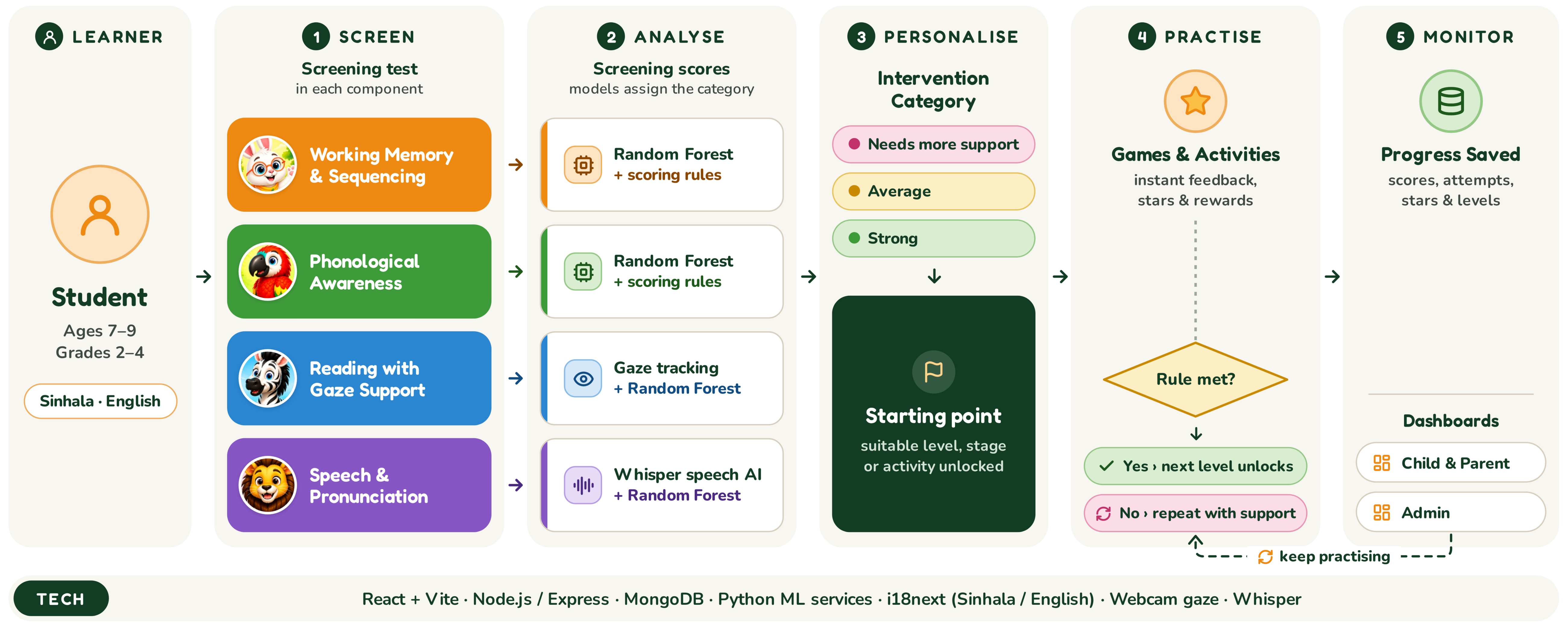
LexiLand learning dashboard

OBJECTIVES

1. **Main:** To develop a bilingual, game-based platform that screens possible dyslexia-related difficulties and provides personalized learning for Sri Lankan students aged 7–9 (Grades 2–4), through a simple, child-friendly and motivating UI/UX design.
2. **Working Memory:** Assess memory, sequencing and recall; provide progressive activities.
3. **Phonological Awareness:** Assess sound, letter and spelling skills in Sinhala and English.
4. **Reading Processing:** Analyze reading accuracy and gaze for adaptive support.
5. **Speech Processing:** Detect pronunciation errors and provide personalized practice.

METHODOLOGY

Data for Working Memory, Phonological Awareness and Speech Processing were **collected manually by conducting a screentest**, while Reading Processing data were obtained from a **Kaggle** dataset. The collected data were cleaned, prepared and used to train the relevant models (**Random Forest** classifiers; **MediaPipe** gaze analysis for reading). Each student completes the screening for a component and is analyzed accordingly; the system then assigns a **support level** and unlocks personalized practice activities (Figure 1). Each component follows its own progression rules and activity structure (Figure 2).



LEARNER
Student
Ages 7–9
Grades 2–4
Sinhala · English

1 SCREEN
Screening test
in each component
Working Memory & Sequencing
Phonological Awareness
Reading with Gaze Support
Speech & Pronunciation

2 ANALYSE
Screening scores
models assign the category
Random Forest + scoring rules
Random Forest + scoring rules
Gaze tracking + Random Forest
Whisper speech AI + Random Forest

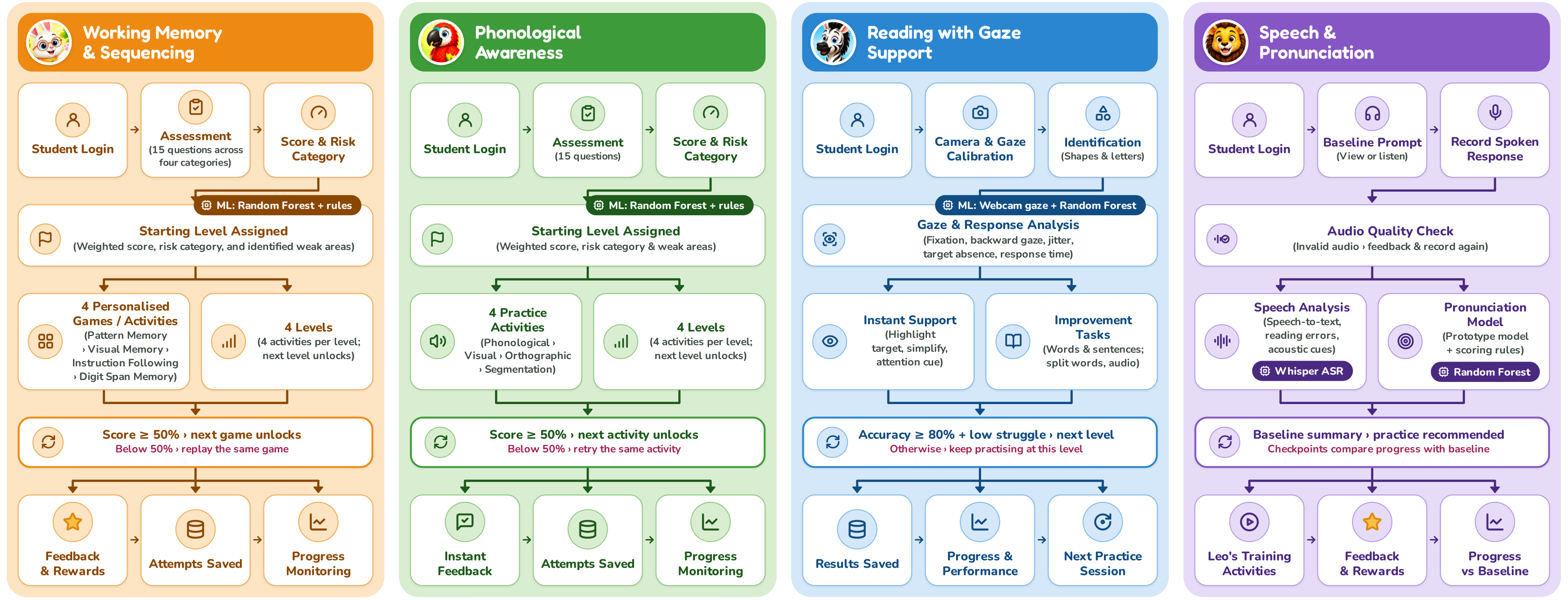
3 PERSONALISE
Intervention Category
● Needs more support
● Average
● Strong
Starting point
suitable level, stage or activity unlocked

4 PRACTISE
Games & Activities
instant feedback, stars & rewards
Rule met?
✓ Yes › next level unlocks
No › repeat with support

5 MONITOR
Progress Saved
scores, attempts, stars & levels
Dashboards
Child & Parent
Admin
keep practising

TECH React + Vite · Node.js / Express · MongoDB · Python ML services · i18next (Sinhala / English) · Webcam gaze · Whisper

Figure 1 · *Overall workflow: Screening → Analysis → Support Level → personalized Practice*



Working Memory & Sequencing
Student Login → Assessment (15 questions across four categories) → Score & Risk Category
ML: Random Forest + rules
Starting Level Assigned (Weighted score, risk category, and identified weak areas)
4 Personalised Games / Activities (Pattern Memory › Visual Memory › Instruction Following › Digit Span Memory)
4 Levels (4 activities per level; next level unlocks)
Score ≥ 50% › next game unlocks
Below 50% › replay the same game
Feedback & Rewards → Attempts Saved → Progress Monitoring

Phonological Awareness
Student Login → Assessment (15 questions) → Score & Risk Category
ML: Random Forest + rules
Starting Level Assigned (Weighted score, risk category & weak areas)
4 Practice Activities (Phonological › Visual › Orthographic › Segmentation)
4 Levels (4 activities per level; next level unlocks)
Score ≥ 50% › next activity unlocks
Below 50% › retry the same activity
Instant Feedback → Attempts Saved → Progress Monitoring

Reading with Gaze Support
Student Login → Camera & Gaze Calibration → Identification (Shapes & letters)
ML: Webcam gaze + Random Forest
Gaze & Response Analysis (Fixation, backward gaze, jitter, target absence, response time)
Instant Support (Highlight target, simplify, attention cue)
Improvement Tasks (Words & sentences; split words, audio)
Accuracy ≥ 80% + low struggle › next level
Otherwise › keep practising at this level
Results Saved → Progress & Performance → Next Practice Session

Speech & Pronunciation
Student Login → Baseline Prompt (View or listen) → Record Spoken Response
Audio Quality Check (Invalid audio › feedback & record again)
Speech Analysis (Speech-to-text, reading errors, acoustic cues) Whisper ASR
Pronunciation Model (Prototype model + scoring rules) Random Forest
Baseline summary › practice recommended
Checkpoints compare progress with baseline
Leo's Training Activities → Feedback & Rewards → Progress vs Baseline

Figure 2 · *Progression structure followed by each component*

UI/UX DESIGN APPROACH

The interface was designed using **dyslexia-friendly color palettes** and design principles, with the palette and overall approach **reviewed and approved by doctors and specialists** working with neurodivergent children. Complete **English and Sinhala** interfaces support bilingual learning, while minimal text, audio guidance and animal guides keep navigation simple.

Color-coded levels
Easy · Medium · Hard · Super

Animal guides

Rewards
stars, feedback & unlockable levels

RESULTS AND DISCUSSION

Feature	Model & feature optimization	Accuracy	Study
Scores + response time → **performance level** → starting game level	Random Forest Balanced class weighting	**94.2%** ± 2.6	This Study Working Memory
scores + response time → **support category** → starting activity level	Random Forest Depth limit + score–time features	**92.8%** ± 3.1	This Study Phonological
44 voice features (pitch, loudness, pauses) → **support need** → speech activities	Random Forest Child-speech Mel + Δ/ΔΔ MFCCs Balanced subsampling + calibration	**84.1%** ± 1.5	This Study Speech
Webcam gaze + responses → **reading difficulty** → instant support	MediaPipe gaze + rules Hyperparameter tuning	**85.6%** ± 11.8	This Study Reading

Evaluation: stratified splitting + 5-fold CV; ± = SD across folds; Speech uses participant-disjoint folds. Δ = change in MFCCs over time; ΔΔ = change in Δ over time.

Discussion. All models exceeded **84% accuracy** (Working Memory highest, 94.2%). In testing, children engaged well; **simplified instructions**, animal guides and **colour-coded levels** improved usability. Results indicate support needs, not a diagnosis.

CONCLUSION

LexiLand integrates memory, sound, reading and speech support in one functional, **bilingual, game-based platform**. High model accuracy enables reliable personalization, and specialist-reviewed UI/UX keeps practice engaging. Future work: validation with larger school samples.

+ + + → **LexiLand**

REFERENCES

1. International Dyslexia Association. (2025). *Definition of dyslexia and explanatory statement.*
2. Marinelli, C. V., et al. (2023). The effectiveness of serious games for enhancing literacy skills in children with learning difficulties. *Applied Sciences, 13*(7), 4512.
3. Peries, W. A. N., et al. (2021). Primary-school teachers' readiness in identifying children with dyslexia: A Sri Lankan study.
4. Weerasinghe, Y., et al. (2025). Gamifying learning for dyslexia in a novel linguistic context.